



南京大學120周年校慶
120th ANNIVERSARY
NANJING UNIVERSITY
1902 - 2022



Academic Lecture Series of Software Institute in Nanjing University

Title: "Demystifying the Cloud Hypes"
Date: April 2, 2024
Time: 2:00 PM
Location: Fei Yimin Building, Room 822
Speaker: Davide Taibi, Full Professor,
University of Oulu (Finland)
Host: Shanshan Li, Assistant Researcher,
Software Institute, Nanjing University



Abstract:

New technologies are constantly introduced in the market. In the Cloud domain, new technologies revolutionised how software is developed. However, while some technologies might be very beneficial in some contexts, they might have a negative impact in others. In our research group, we apply empirical methods to demystify hypes in new cloud-related technologies, including microservices, serverless, edge, and quantum computing. Moreover, we are actively researching **methods to detect and reduce architectural degradation of cloud-based systems**, proposing **new metrics, methods, and visualisations** to help companies to **keep the technical debt of cloud-based applications** under control.

Short Bio:

Davide Taibi is a Full Professor at the University of Oulu (Finland). His research is mainly focused on the evolution and assessment of cloud-native systems, with a particular focus on the migration from monolithic to cloud-native applications. He is investigating processes and techniques for developing Cloud Native applications, identifying approaches to decompose monolithic systems into microservices and detecting architectural degradation or cloud-native systems. Moreover, he has a deep experience in assessing the quality of software systems, with a special focus on Technical Debt and Code Quality Assessment. He has been a member of the International Software Engineering Network (ISERN) since 2018. Before moving to Finland, he has been Assistant Professor at the Free University of Bozen/Bolzano (2015-2017), a post-doctoral research fellow at the Technical University of Kaiserslautern and Fraunhofer Institute for Experimental Software Engineering - IESE (2013-2014), and research fellow at the University of Insubria (2007-2011).





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Academic Lecture Series of Software Institute in Nanjing University

Title: "Technical Debt: The Dark Side of Developing"

Date: April 2, 2024

Time: 3:00 PM

Location: Fei Yimin Building, Room 822

Speaker: Valentina Lenarduzzi,
Assistant Professor,
University of Oulu (Finland)

Host: Shanshan Li, Assistant Researcher,
Software Institute, Nanjing University



Abstract:

Software is always evolving. Developers commonly postpone maintenance, accumulating the so called Technical Debt (TD). In the last few years, the number of publications on TD is constantly increasing. Moreover, several companies started to adopt TD analysis tools such as SonarQube and Cast, investing a relevant amount of budget on refactoring activities recommended by these tools, with the hope of reducing bugs or ease the software maintenance. This is certainly a very encouraging sign, where software engineering research topics receive balanced attention by both communities. Several researchers also proposed methods to estimate TD. Unfortunately, there is still a gap on the identification of the relationship between TD and software qualities and how to determine the impact of TD on software quality. In this presentation, we will go through the stages of bad and best practices for how to manage TD.

Short Bio:

Valentina Lenarduzzi is an Assistant Professor (tenure track) at the University of Oulu (Finland). She is a Finnish consortium and WG5 leader in the COST Action EUGAIN European Network For Gender Balance in Informatics. Her research activities are related to contemporary software development practices and methodologies, including data analysis in software engineering, software quality, software maintenance, and evolution, focusing on Technical Debt as well as code and architectural smells. She got her Ph.D. in Computer Science in 2015 and was a postdoctoral researcher at the Free University of Bozen-Bolzano, (Italy), Tampere University (Finland), and LUT University (Finland). Moreover, she was visiting researcher at the University of Kaiserslautern (TUK) and the Fraunhofer Institute for Experimental Software Engineering IESE (Germany). She served as a program committee member of various international conferences (e.g., ICPC, ICSME, ESEM), and for various international journals (e.g., TSE, EMSE, JSS, IST) in the field of software engineering. She has been program co-chair of OSS 2021 and TechDebt 2022. She was also one of the organizers of the last edition of MaLTesQuE workshop (2022) collocated with ES-EC/FSE. Dr. Lenarduzzi is recognized by the Journal of Systems and Software (JSS) as one of the most active SE researchers in top-quality journals in the period 2013 to 2020.

